

Scenario for Project 2

Heaven Classics successfully creates an EC2 Server Instance for Windows 2012 Server. After launching the instance on the server, the next step was to monitor the operations.

Monitoring is important to keep an eye on the performance of an EC2 instance. It helps gather data from all parts, and is useful for debugging failure.

The monitoring team at Heaven Classics started monitoring activities using the CloudWatch Service in the AWS Management Console. The Heaven Classics support team were required to meet the following objectives:

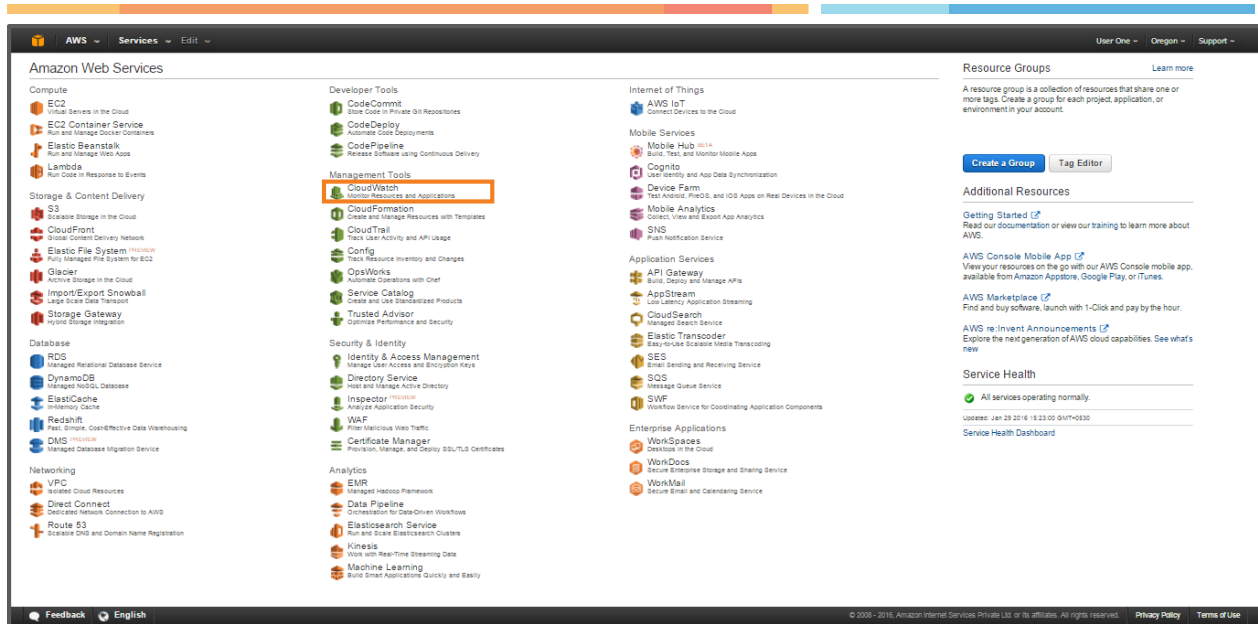
1. Check the CPU Utilization.
2. Create an Alarm.
3. Create an IAM User.
4. Create the IAM Administrator Group, and add the user to the Administrator Group.
5. Create a Role.

1. Check the CPU Utilization

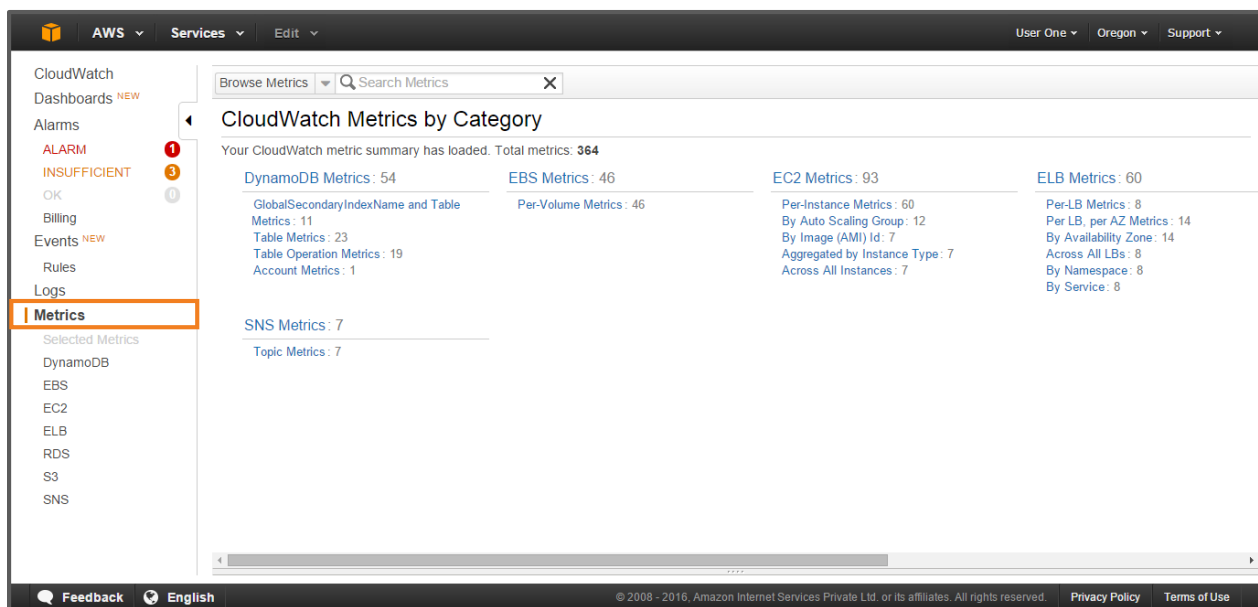
After deciding the monitoring goals, the next step is to decide the minimum usage for an EC2 instance, and monitoring it.

To check the CPU Utilization:

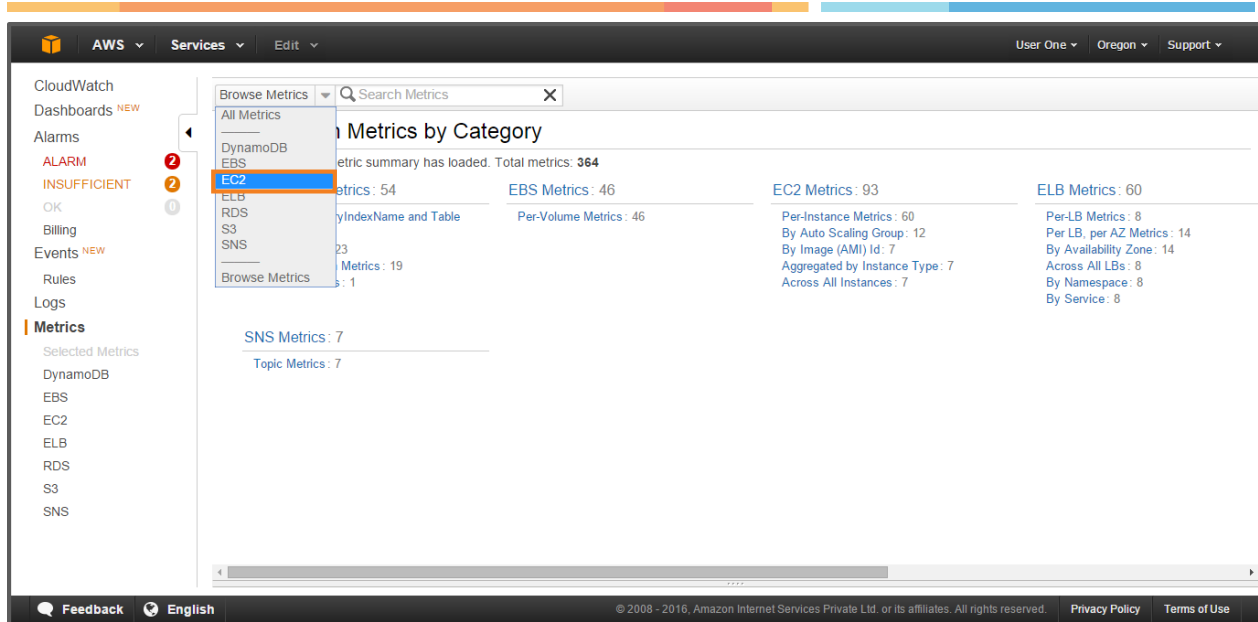
1. Sign in to the AWS Management Console to open the CloudWatch Console.



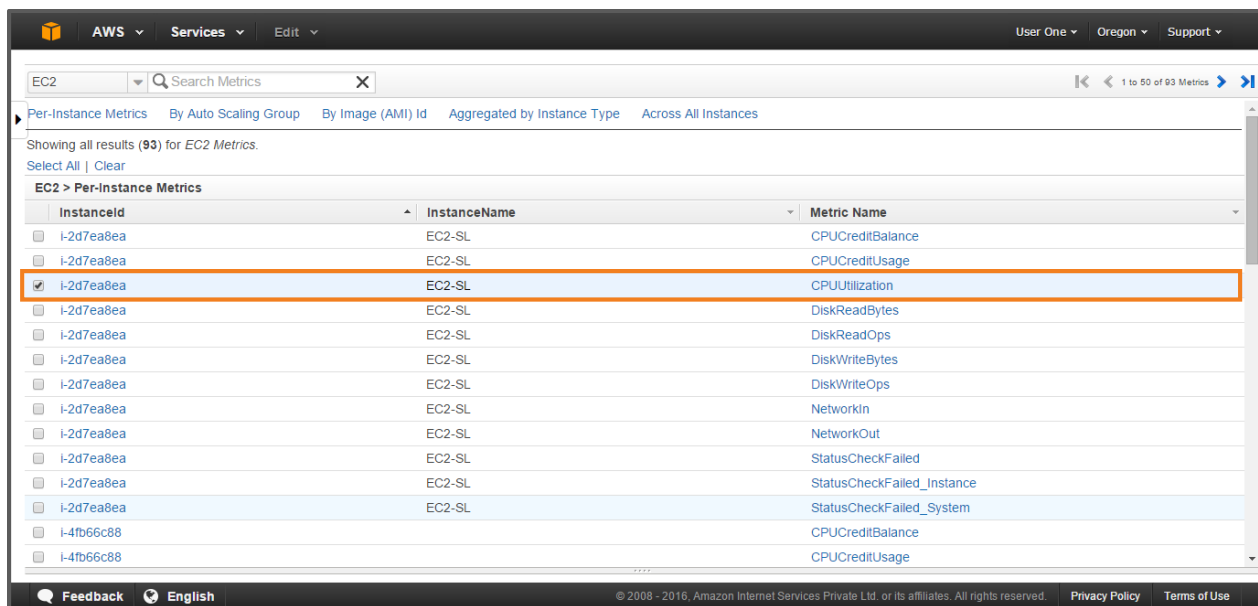
2. On the Navigation pane, click **Metrics**.



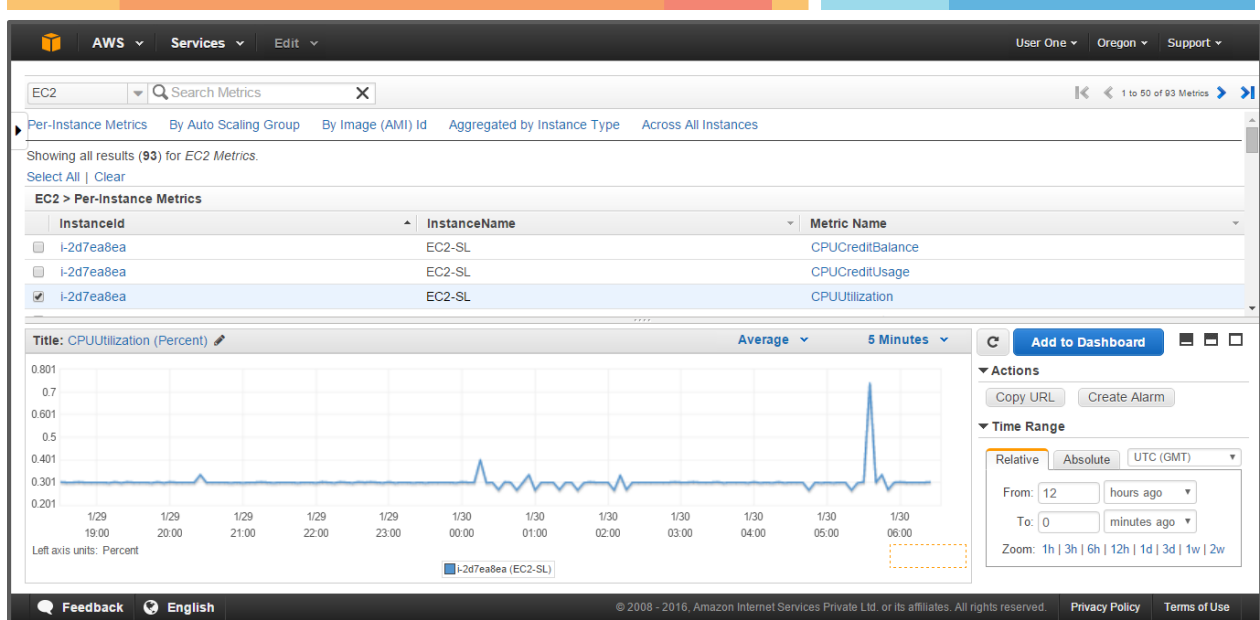
3. Then, in the **CloudWatch Metrics by Category** pane, select the **EC2 Metrics** category from the Browse Metrics drop down list.



4. Next, select a row of metric that contains **CPUUtilization**.



5. After selecting CPUUtilization, look for the graph that presents average CPUUtilization for a single instance in the details pane. This shows the CPU utilization. The monitoring team can review this graph while monitoring the system resources.

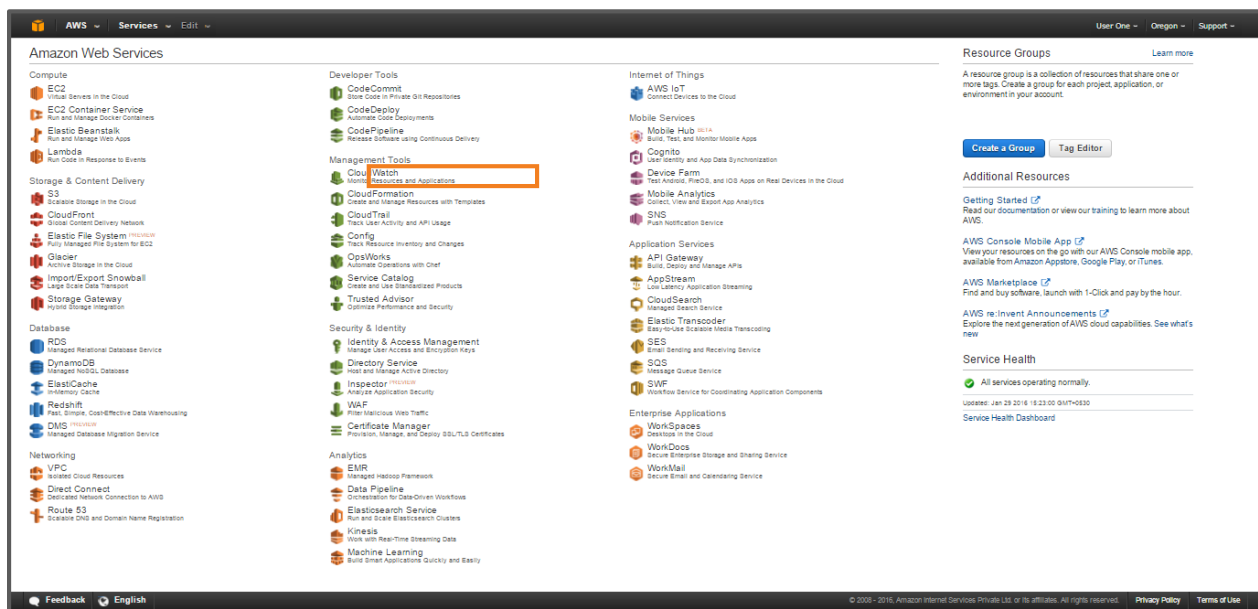


2. Create an Alarm

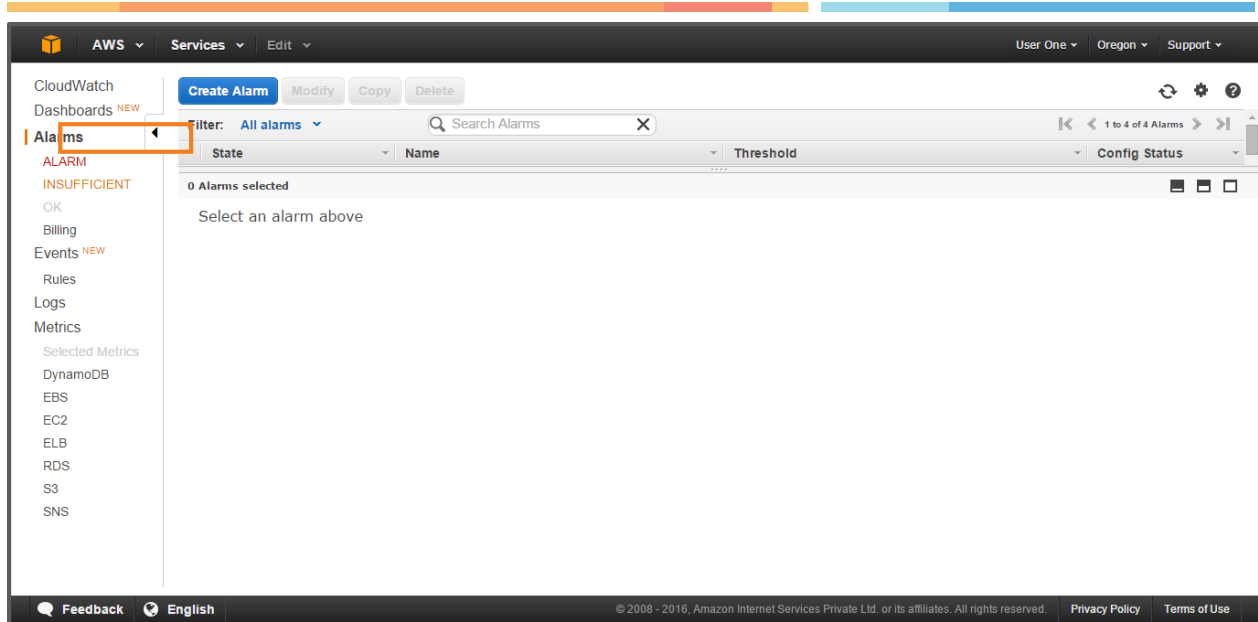
Along with monitoring the CPU utilization, the support team sets an alarm to send notifications to the monitoring team. This helps them in quick tracking the current state of the system, and taking appropriate action if needed. Let's now see the steps to create an alarm.

To create an Alarm:

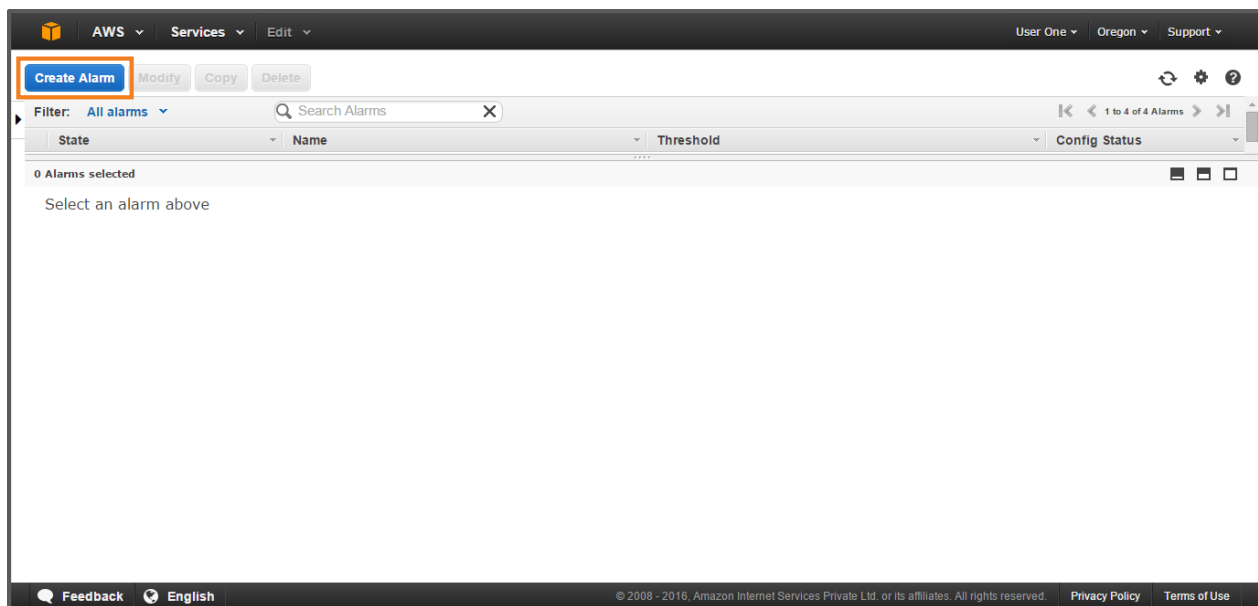
1. Sign in to the AWS Management Console to open the CloudWatch Console.



2. On the Navigation pane, click **Alarms**.



3. Then, click **Create Alarm**.



4. In the **Create Alarm** pane, select the **EC2 Metrics** category from the Browse Metrics drop down list.

Create Alarm

1. Select Metric

2. Define Alarm

EC2

Search Metrics

1 to 50 of 117 Metrics

All Metrics

DynamoDB

EBS

EC2

ELB

RDS

S3

SNS

Browse Metrics

By Auto Scaling Group

By Image (AMI) Id

Aggregated by Instance Type

Across All Instances

Update Graph

Time Range

Relative

Absolute

UTC (GMT)

From: 12

hours ago

To: 0

minutes ago

Zoom: 1h | 3h | 6h | 12h | 1d | 3d | 1w | 2w

Click a metric above to view graph

Click a checkbox to select a metric

Click on text to add to search

Cancel

Previous

Next

Create Alarm

5. Next, select the metric name, CPUUtilization.

Create Alarm

1. Select Metric

2. Define Alarm

EC2

Search Metrics

1 to 50 of 117 Metrics

Per-Instance Metrics

By Auto Scaling Group

By Image (AMI) Id

Aggregated by Instance Type

Across All Instances

EC2 > Per-Instance Metrics

InstanceID	InstanceName	Metric Name
☐ i-2d7ea8ea	EC2-SL	CPUCreditBalance
☐ i-2d7ea8ea	EC2-SL	CPUCreditUsage
☒ i-2d7ea8ea	EC2-SL	CPUUtilization
☐ i-2d7ea8ea	EC2-SL	DiskReadBytes
☐ i-2d7ea8ea	EC2-SL	DiskReadOps
☐ i-2d7ea8ea	EC2-SL	DiskWriteBytes

Title: CPUUtilization (Percent)

Average

5 Minutes

Update Graph

Time Range

Relative

Absolute

UTC (GMT)

From: 12

hours ago

To: 0

minutes ago

0.801

0.601

0.401

0.201

1/29 20:00

1/29 22:00

1/30 00:00

1/30 02:00

1/30 04:00

1/30 06:00

Left axis units: Percent

i-2d7ea8ea (EC2-SL)

Cancel

Previous

Next

Create Alarm

6. Then, click **Next**.

Create Alarm

1. Select Metric

2. Define Alarm

EC2

Search Metrics

1 to 50 of 117 Metrics

Per-Instance Metrics

By Auto Scaling Group

By Image (AMI) Id

Aggregated by Instance Type

Across All Instances

EC2 > Per-Instance Metrics

InstanceID	InstanceName	Metric Name
☐ i-2d7ea8ea	EC2-SL	CPUCreditBalance
☐ i-2d7ea8ea	EC2-SL	CPUCreditUsage
☒ i-2d7ea8ea	EC2-SL	CPUUtilization
☐ i-2d7ea8ea	EC2-SL	DiskReadBytes
☐ i-2d7ea8ea	EC2-SL	DiskReadOps
☐ i-2d7ea8ea	EC2-SL	DiskWriteBytes

Title: CPUUtilization (Percent)

Average

5 Minutes

Update Graph

Time Range

Relative

Absolute

UTC (GMT)

From: 12

hours ago

To: 0

minutes ago

0.801

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1/29 20:00

1/29 22:00

1/30 00:00

1/30 02:00

1/30 04:00

1/30 06:00

Left axis units: Percent

i-2d7ea8ea (EC2-SL)

Cancel

Previous

Next

Create Alarm

- In the **Alarm Threshold** section, enter the fields: **Name:** Alarm1, **Description:** This is Alarm1, **is:** ≤ 3 , **for:** 3 consecutive periods(s). In **Actions** section, select the type of action as: **Whenever this alarm:** State is ALARM, **Send notification to:** Event1, **Email list:** HC_Monitor@HeavenClassics.com, **Period:** 5 Minutes.

Create Alarm

1. Select Metric

2. Define Alarm

Alarm Threshold

Provide the details and threshold for your alarm. Use the graph on the right to help set the appropriate threshold.

Name:

Description:

Whenever: CPUUtilization

is:

for: consecutive period(s)

Alarm Preview

This alarm will trigger when the blue line goes down to or below the red line for a duration of 15 minutes

CPUUtilization <= 3

Namespace: AWS/EC2

InstanceId:

InstanceName: EC2-SL

Metric Name:

Period:

Statistics:

Actions

Define what actions are taken when your alarm changes state.

Notification

Delete

Whenever this alarm:

Send notification to: [New list](#) [Enter list](#) ⓘ

Email list:

Cancel

Previous

Next

Create Alarm

8. Next, click **Create Alarm**.

Create Alarm

1. Select Metric

2. Define Alarm

Alarm Threshold

Provide the details and threshold for your alarm. Use the graph on the right to help set the appropriate threshold.

Name:

Alarm1

Description:

This is Alarm1

Whenever:

CPUUtilization

is:

<=

3

for:

3

consecutive period(s)

Alarm Preview

This alarm will trigger when the blue line goes down to or below the red line for a duration of 15 minutes

CPUUtilization <= 3

Namespace:

AWS/EC2

InstanceId:

i-2d7ea8ea

InstanceName:

EC2-SL

Metric Name:

CPUUtilization

Period:

5 Minutes

Statistics:

Cancel

Previous

Next

Create Alarm

9. The Alarm is created.

AWS

Services

Edit

User One Oregon Support

✓

Your alarm Alarm1 has been saved.

Create Alarm

Modify

Copy

Delete

Filter: All alarms

Search Alarms

State	Name	Threshold	Config Status
ALARM	13	CPUUtilization <= 3 for 2 minutes	Pending confirmation
ALARM	Alarm1	CPUUtilization <= 3 for 15 minutes	Pending confirmation
INSUFFICIENT_DATA	Books-ReadCapacityUnitsLimit-BasicAlarm	ConsumedReadCapacityUnits >= 240 for 12 minutes	No notifications
INSUFFICIENT_DATA	Purchase-ReadCapacityUnitsLimit-BasicAlarm	ConsumedReadCapacityUnits >= 240 for 12 minutes	No notifications

1 Alarm selected

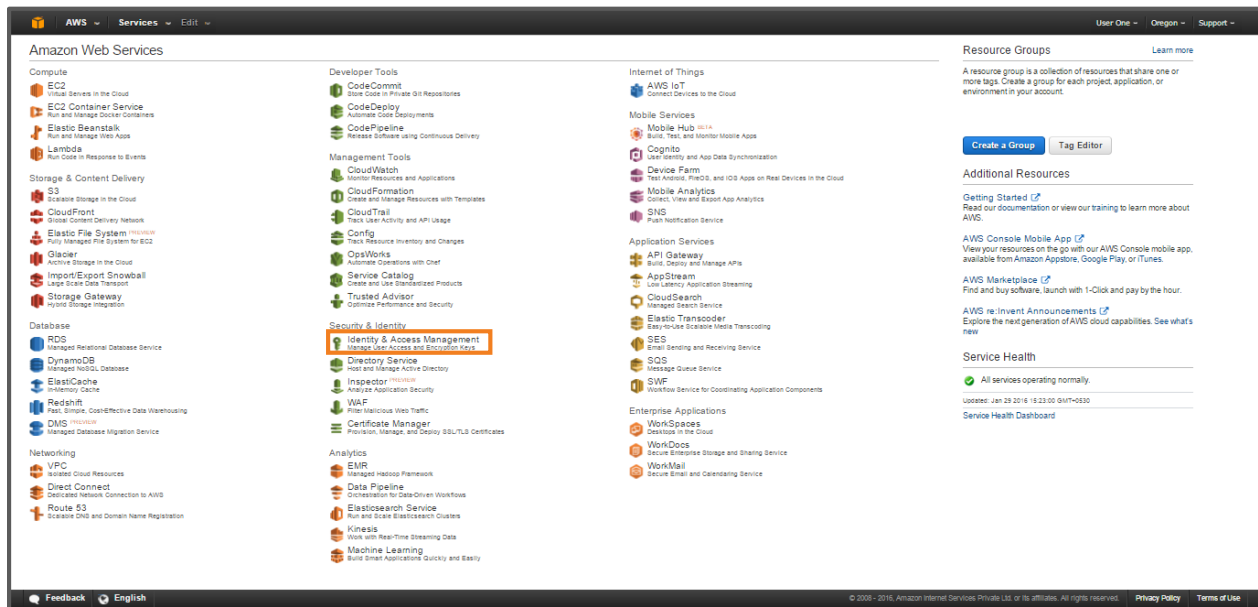
Alarm:Alarm1

3. Create an IAM User

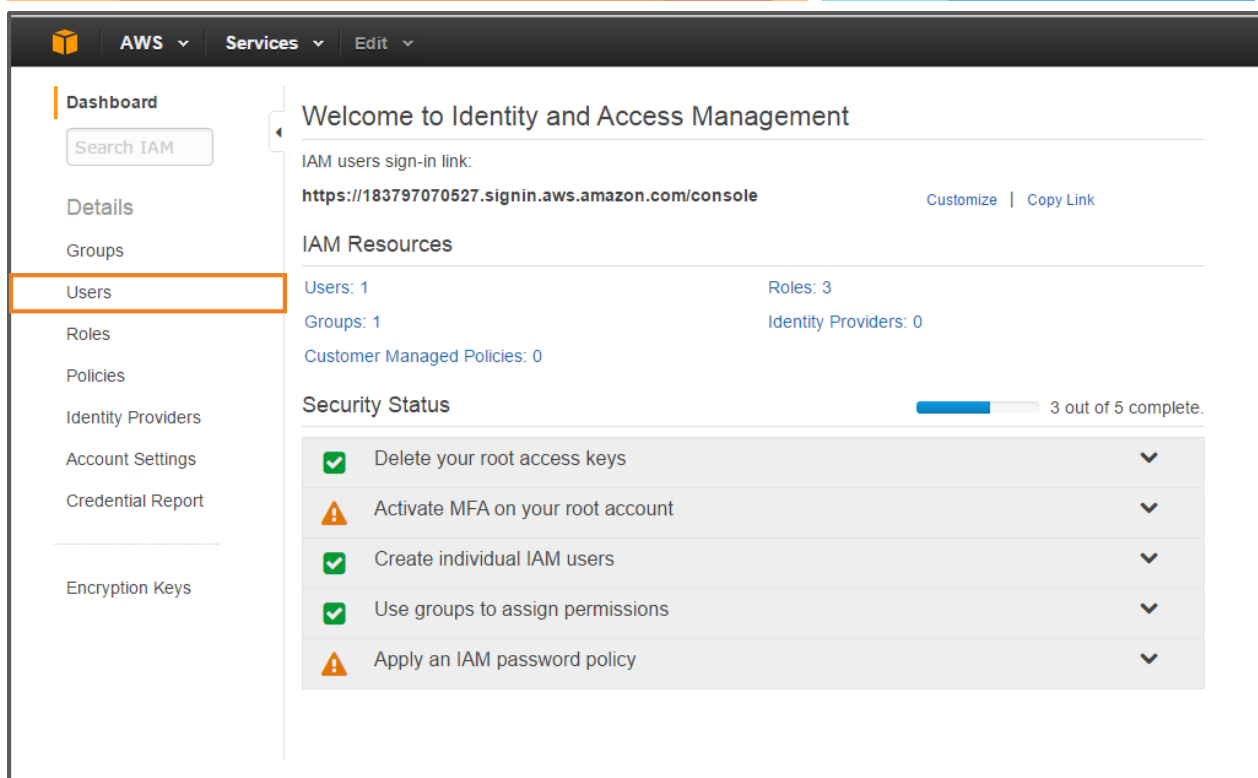
Eventually, the company decided to add employees as Users in the AWS account. Amazon allows to create one or more IAM users in a single AWS account. The support team added existing and new employees as AWS users.

To create an AMI User:

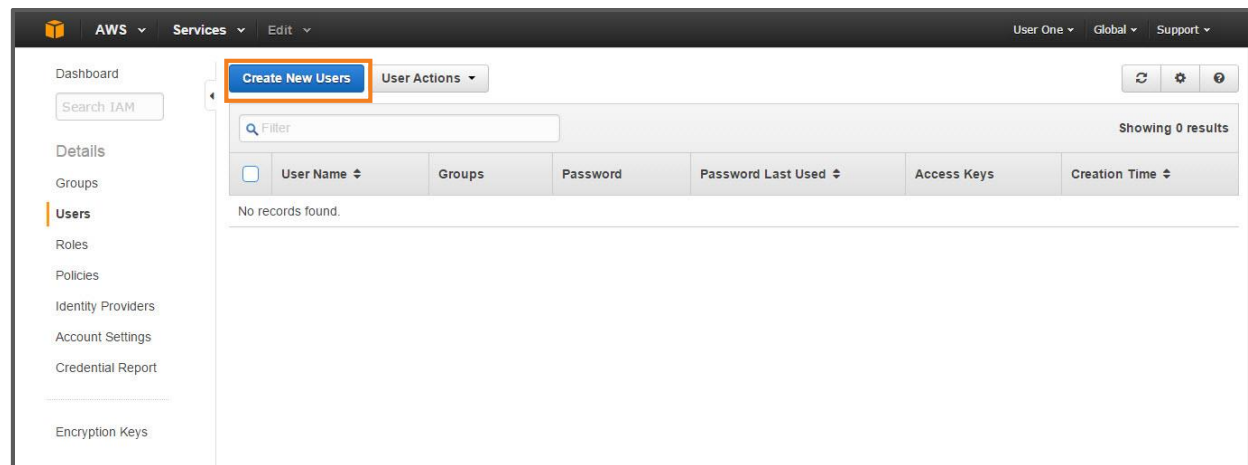
1. Sign in to the AWS Management Console, and open the IAM Console.



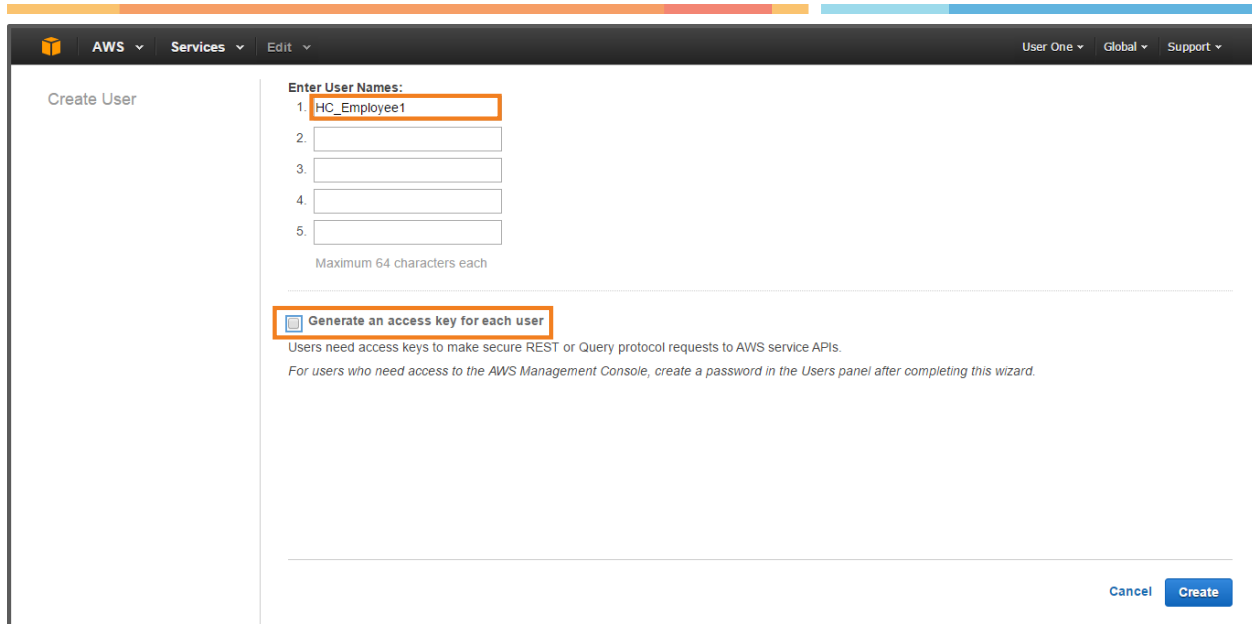
2. On the navigation pane, click **Users**.



3. Then, click **Create New User**.



4. In the Create User wizard, in box 1, enter the **User Name** as HC_Employee1. Similarly, you can enter a list of User Names. Next, clear the check box next to **Generate an access key for each user**.



Create User

Enter User Names:

1. HC_Employee1
- 2.
- 3.
- 4.
- 5.

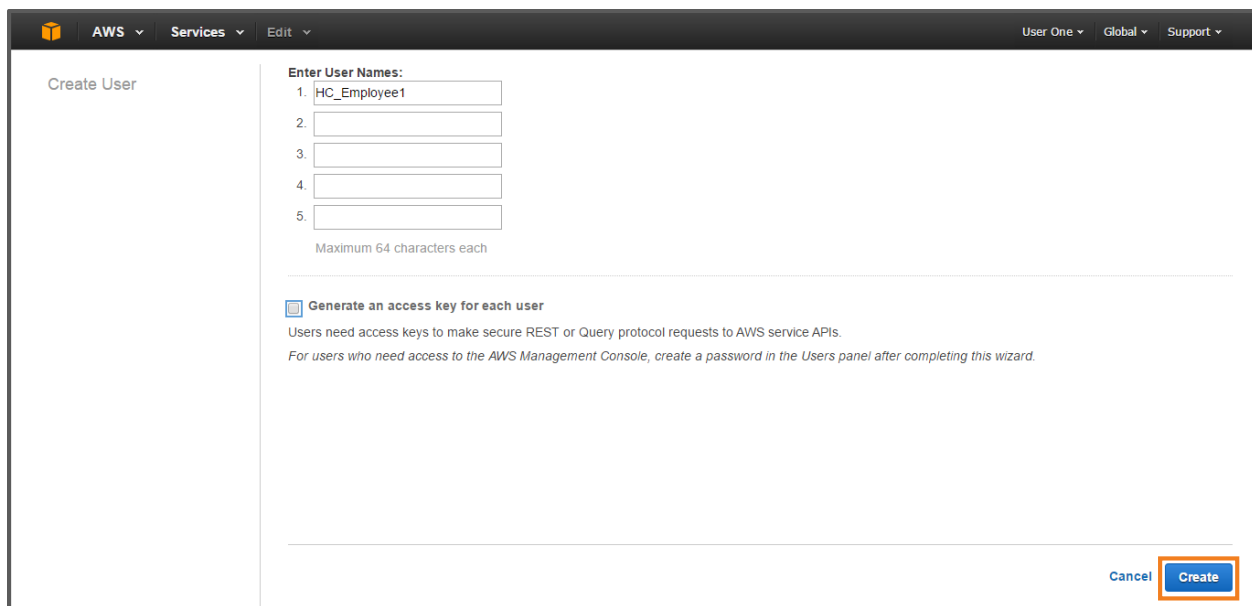
Maximum 64 characters each

☒ Generate an access key for each user

Users need access keys to make secure REST or Query protocol requests to AWS service APIs.
For users who need access to the AWS Management Console, create a password in the Users panel after completing this wizard.

Cancel Create

5. Finally, click **Create**.



Create User

Enter User Names:

1. HC_Employee1
- 2.
- 3.
- 4.
- 5.

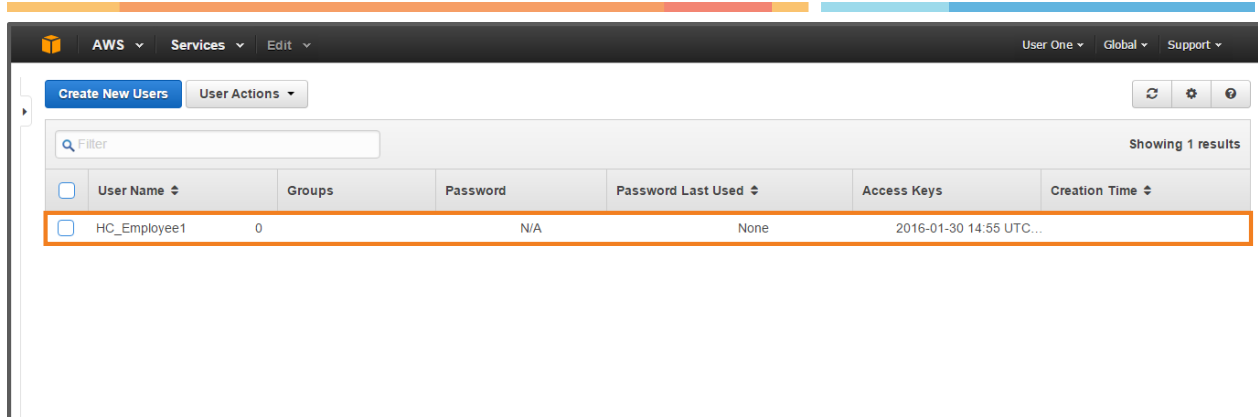
Maximum 64 characters each

☒ Generate an access key for each user

Users need access keys to make secure REST or Query protocol requests to AWS service APIs.
For users who need access to the AWS Management Console, create a password in the Users panel after completing this wizard.

Cancel Create

6. An IAM User is created.

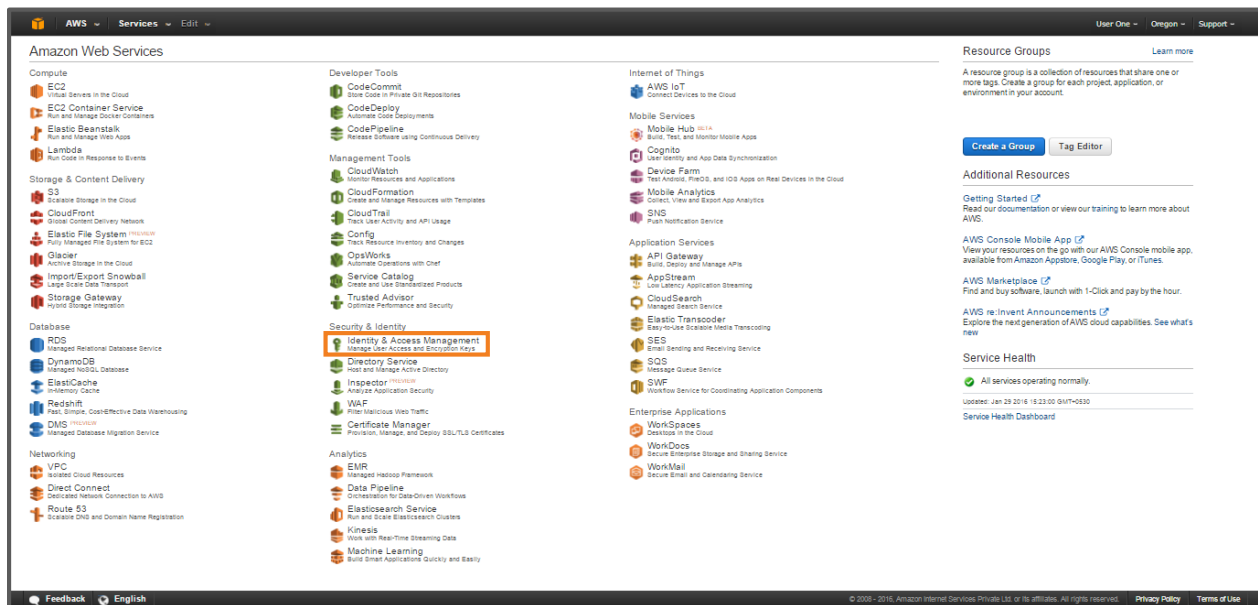


4. Create the IAM Administrator Group, and add the user to the Administrator Group

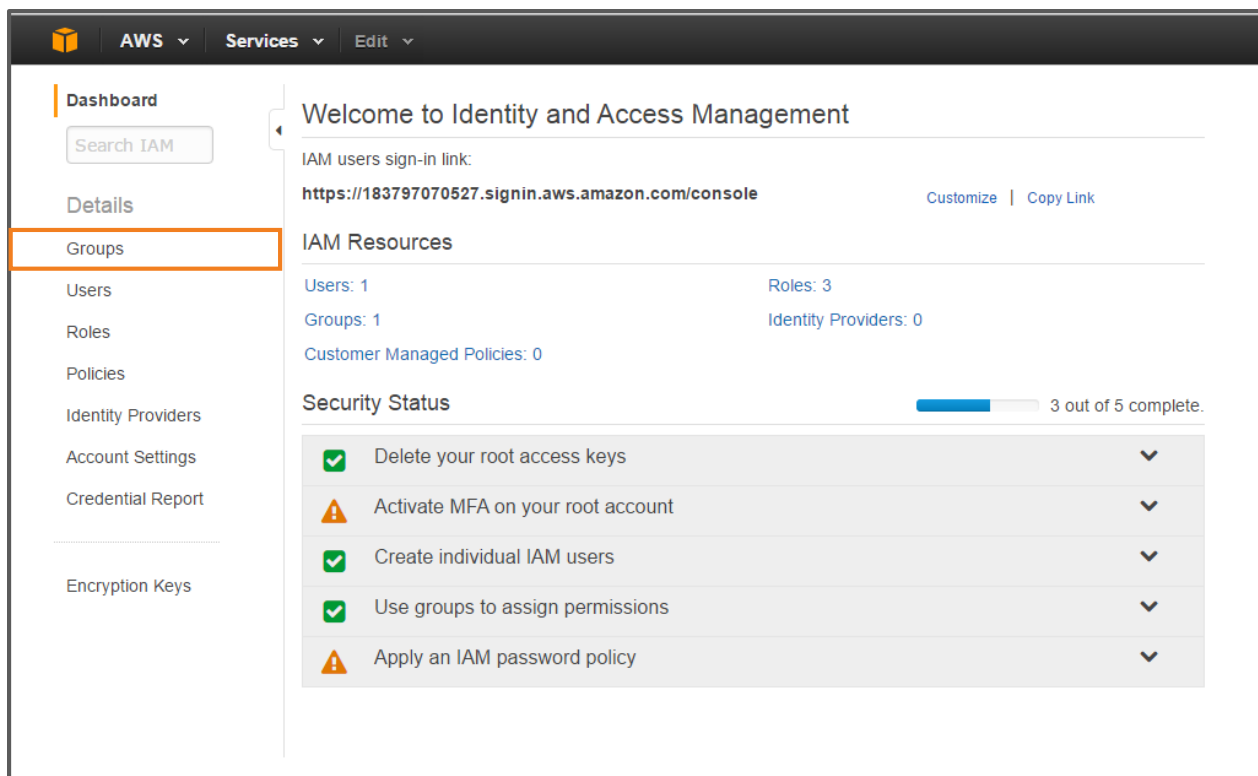
After creating IAM Users, the company formed the IAM Administrator Group. Grouping helped the company to grant a particular permission or allow access for the specified group only.

To create the Administrator Group:

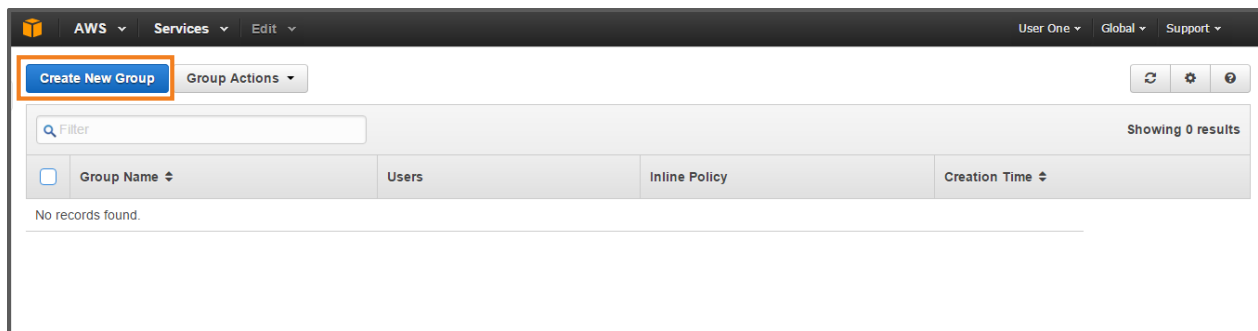
1. Sign in to the AWS Management Console, and open the IAM console.



2. On the Navigation pane, click **Groups**.



3. Then, click **Create New Group**.



4. In the **Create New Group Wizard**, set the **Group Name** in the given text box as: **Administrator**. Then, click **Next Step**.

Create New Group Wizard

Step 1: Group Name

Step 2: Attach Policy

Step 3: Review

Set Group Name

Specify a group name. Group names can be edited any time.

Group Name: **Administrator**

Example: Developers or ProjectAlpha
Maximum 128 characters

Cancel **Next Step**

5. Next, assign the policy name as **AdministratorAccess**, and click **Next Step**.

Create New Group Wizard

Step 1: Group Name

Step 2: Attach Policy

Step 3: Review

Attach Policy

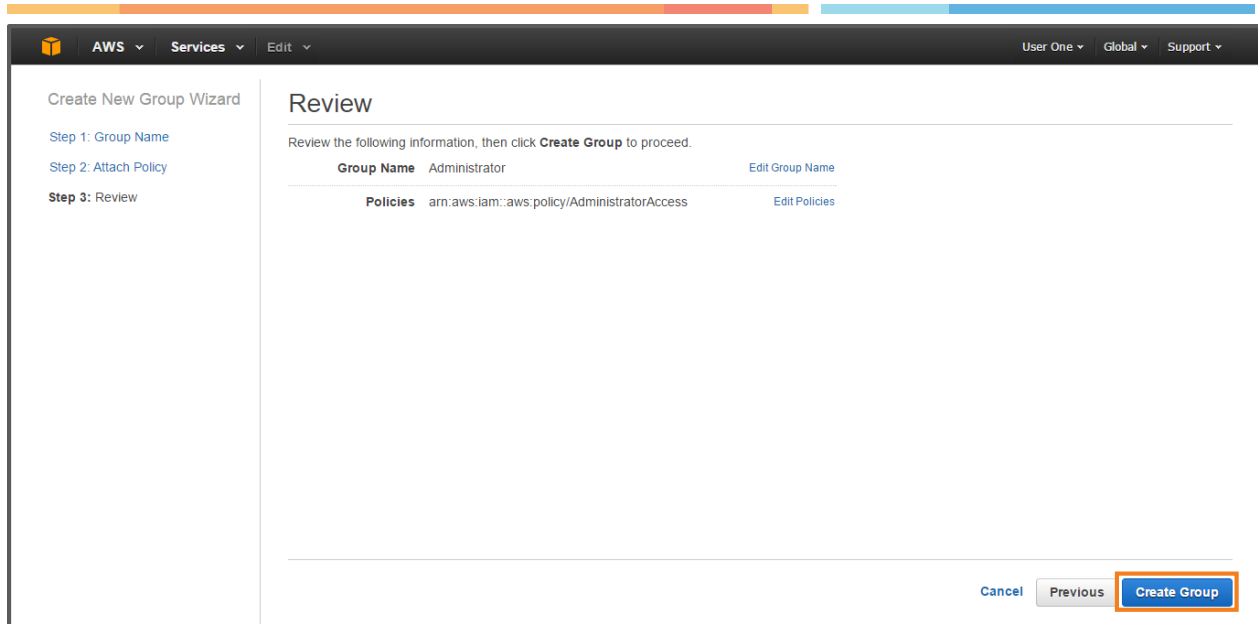
Select one or more policies to attach. Each group can have up to 10 policies attached.

Filter: Policy Type Showing 187 results

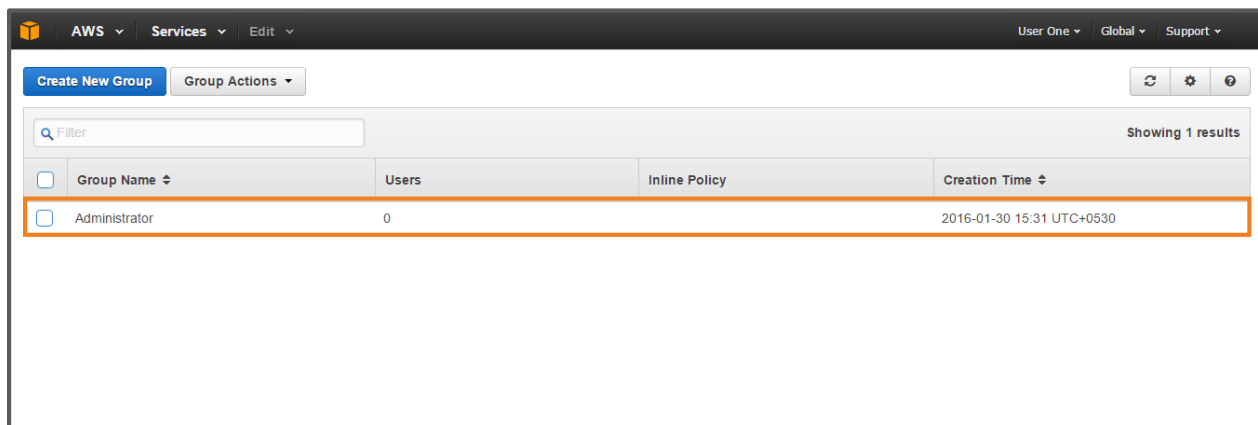
	Policy Name	Attached Entities	Creation Time	Edited Time
☒	AdministratorAccess	2	2015-02-07 00:09 UTC+0530	2015-02-07 00:09 UTC+0530
☐	AmazonAPIGatewayAdminstr...	0	2015-07-09 23:04 UTC+0530	2015-07-09 23:04 UTC+0530
☐	AmazonAPIGatewayInvokeFul...	0	2015-07-09 23:06 UTC+0530	2015-07-09 23:06 UTC+0530
☐	AmazonAPIGatewayPushToCL...	0	2015-11-12 05:11 UTC+0530	2015-11-12 05:11 UTC+0530
☐	AmazonAppStreamFullAccess	0	2015-02-07 00:10 UTC+0530	2015-02-07 00:10 UTC+0530
☐	AmazonAppStreamReadOnlyA...	0	2015-02-07 00:10 UTC+0530	2015-02-07 00:10 UTC+0530
☐	AmazonCognitoDeveloperAuth...	0	2015-03-24 22:52 UTC+0530	2015-03-24 22:52 UTC+0530
☐	AmazonCognitoPowerUser	0	2015-03-24 22:44 UTC+0530	2015-03-24 22:44 UTC+0530
☐	AmazonCognitoReadOnly	0	2015-03-24 22:36 UTC+0530	2015-03-24 22:36 UTC+0530

Cancel Previous **Next Step**

6. Then, review the details of your group request. Finally, click **Create Group**.



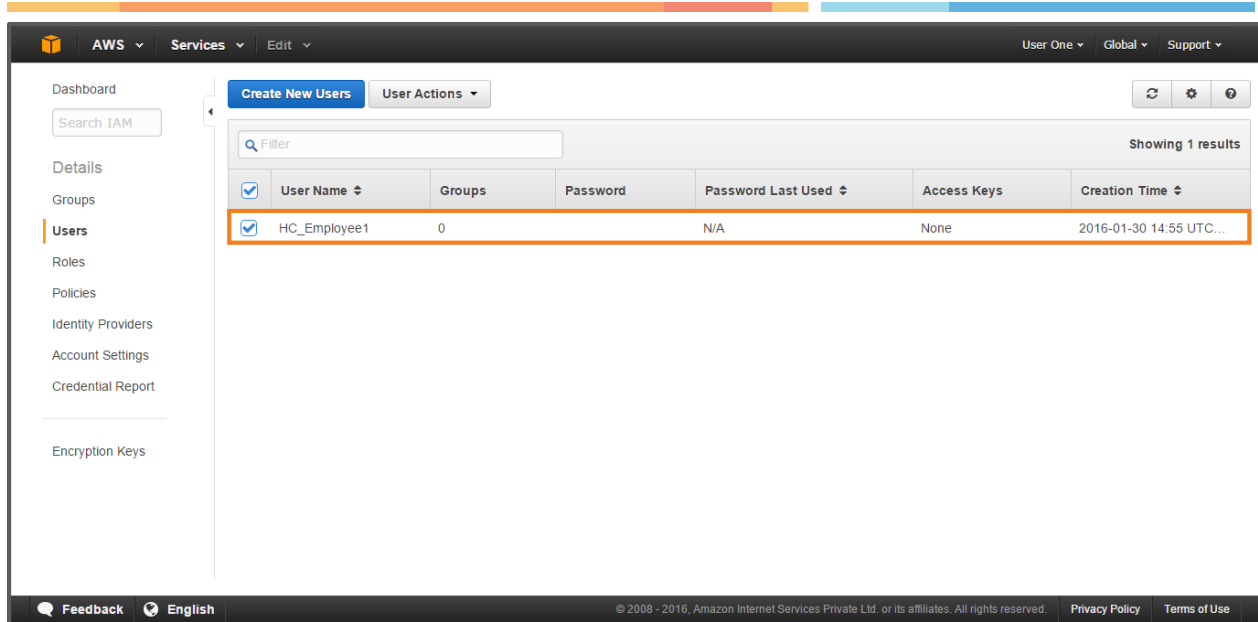
7. A Group is created.



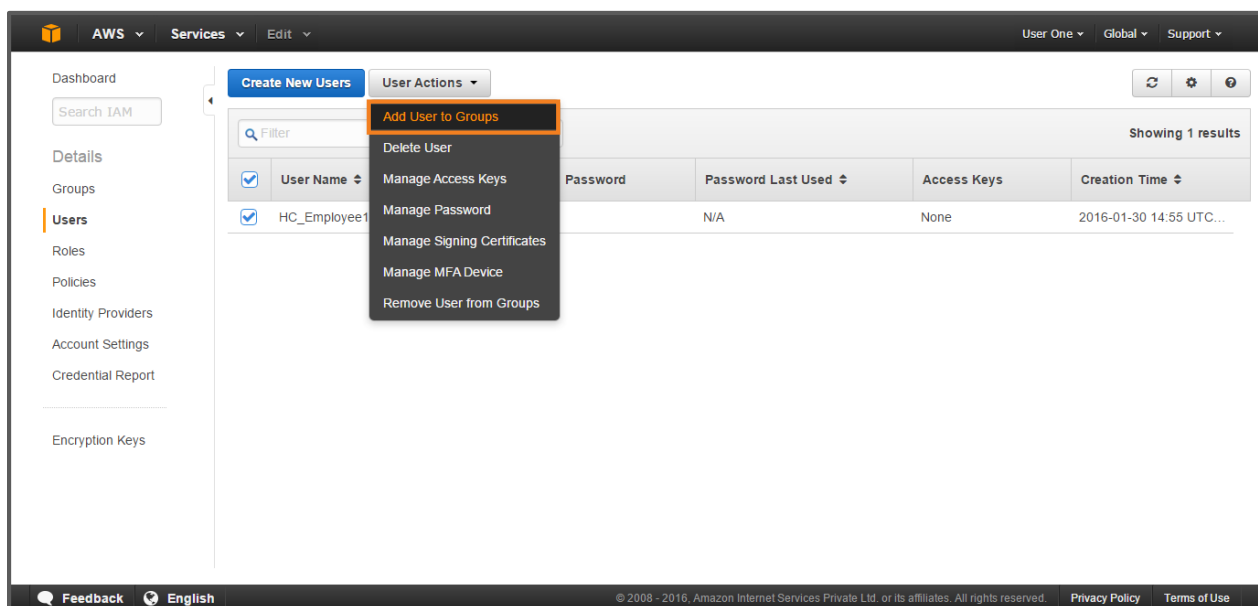
Once the group is created, the team needs to identify the users to be assigned to it. To do this, the team has to add users to the group.

To add Users into the Administrator Group:

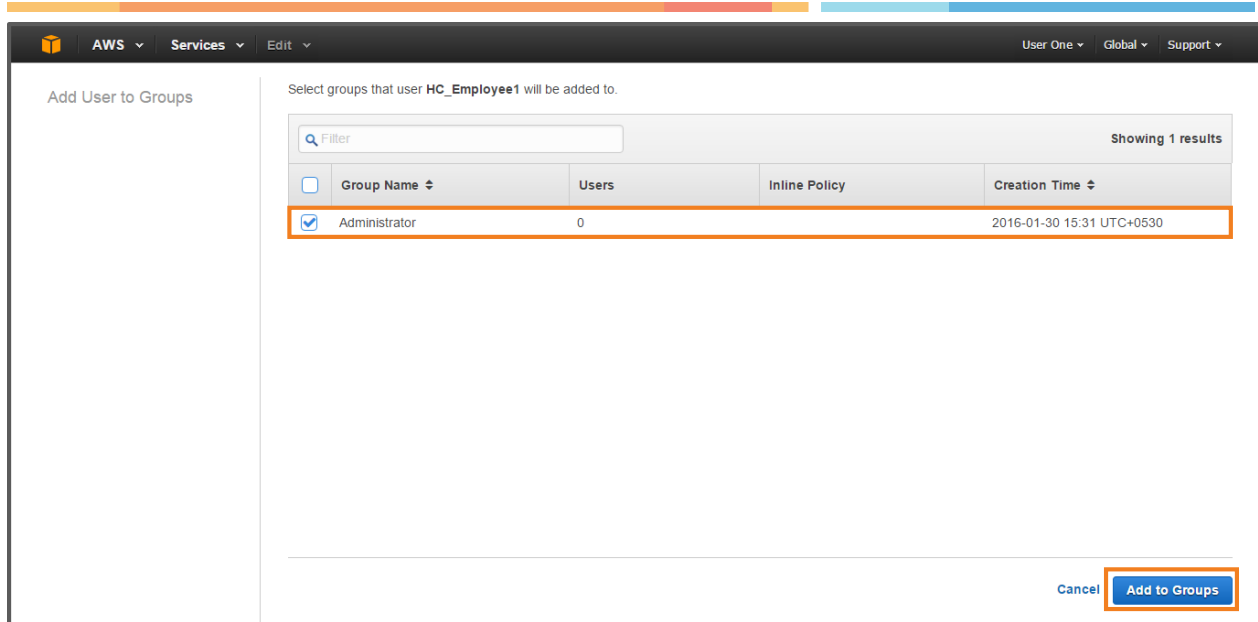
8. From the IAM Console, select the user name to add to the group.
Select **HC_Employee1**.



9. Click **User Action** drop down, and then click **Add User to Groups**.



10. Here, select the group that needs to be added. Then, click **Add to Groups**.

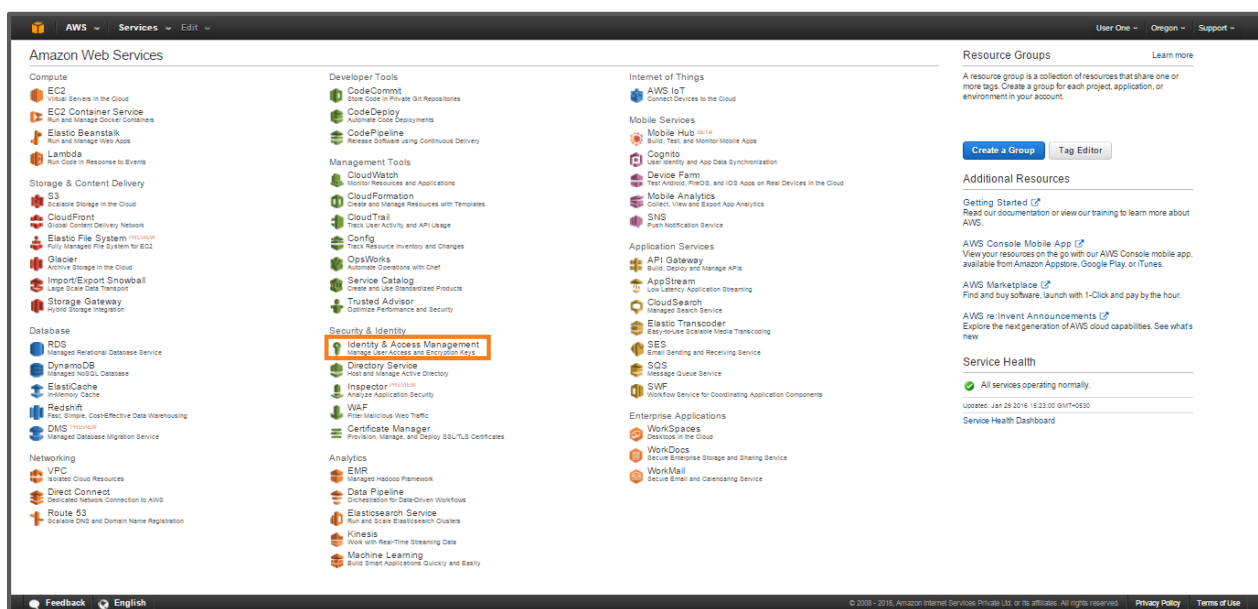


5. Create a Role

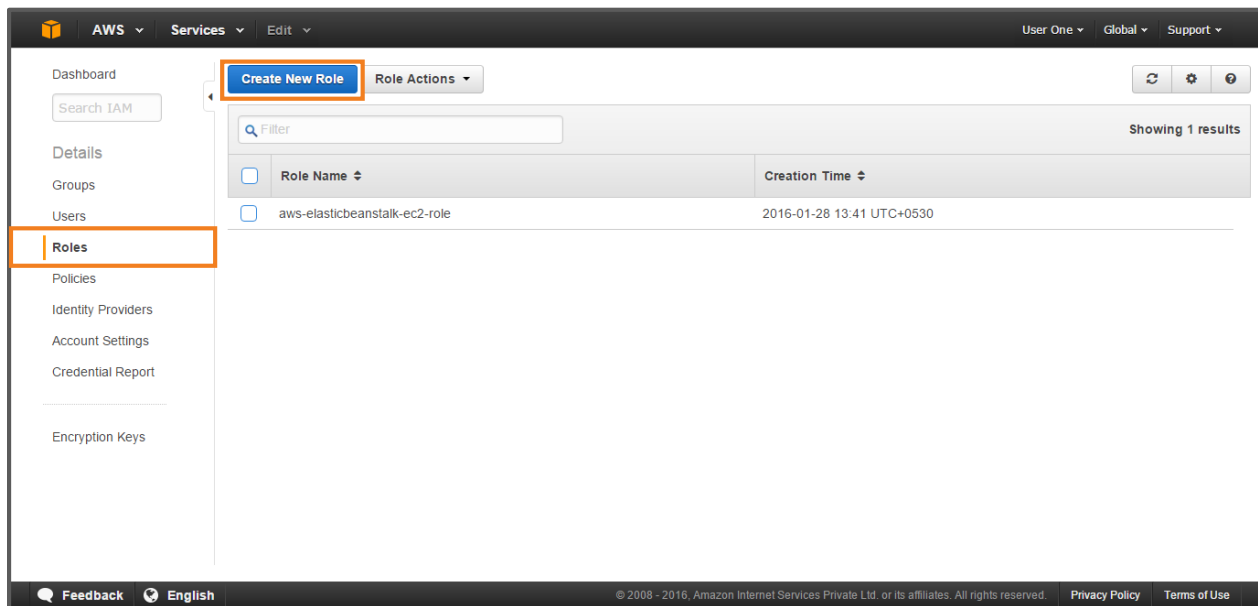
Creating a Role allows access to the AWS resources. This process helps the team to manage user access.

To create a Role:

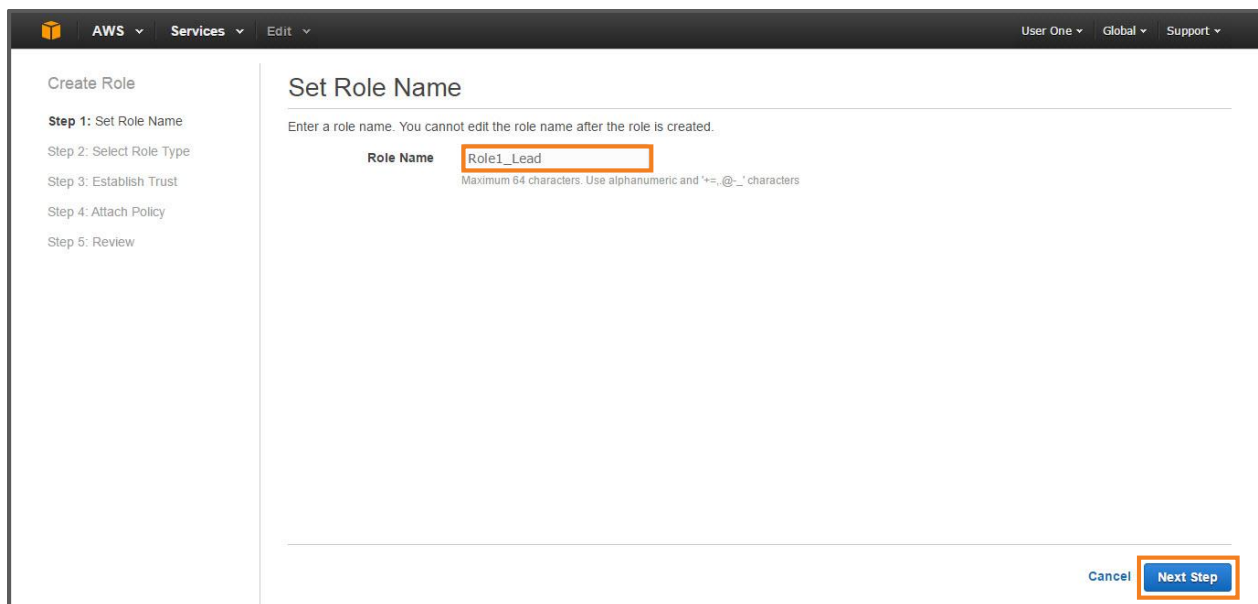
1. Sign in to the AWS Management Console to open the IAM console.



2. On the Navigation pane, click **Roles**. Then, click **Create New Role**.



3. In the **Create Role** wizard, type the **Role Name** in the given text box as: **Role1_Lead**. Then, click **Next Step**.



- The **Select Role Type** page opens. Here, select **Role for Cross-Account Access** section under **AWS Service Roles**, and then select the option as **Provide access between AWS accounts you own**.

The screenshot shows the 'Select Role Type' page in the AWS IAM console. On the left, a sidebar lists the steps: 'Step 1: Set Role Name', 'Step 2: Select Role Type' (current), 'Step 3: Establish Trust', 'Step 4: Attach Policy', and 'Step 5: Review'. The main content area is titled 'Select Role Type'. It has two main sections: 'AWS Service Roles' and 'Role for Identity Provider Access'. Under 'AWS Service Roles', the 'Role for Cross-Account Access' option is selected and highlighted with an orange border. Below this, there are two sub-options: 'Provide access between AWS accounts you own' (which is also highlighted with an orange box) and 'Allows IAM users from a 3rd party AWS account to access this account.'. Each sub-option has a 'Select' button. At the bottom right, there are 'Cancel', 'Previous', and 'Next Step' buttons, with 'Next Step' highlighted by an orange box.

- Next, specify the AWS **Account ID** to grant access to your resources. Enter the ID of created account: 183797070527. Then, check the **Require MFA** box, and click **Next Step**.

The screenshot shows the 'Establish Trust' page in the AWS IAM console. The sidebar shows 'Step 3: Establish Trust' as the current step. The main content area has a heading 'Enter the ID of the AWS account whose IAM users will be able to access this account.' Below this, there is an 'Account ID' field containing the value '183797070527' and a 'Require MFA' checkbox that is checked. At the bottom right, there are 'Cancel', 'Previous', and 'Next Step' buttons, with 'Next Step' highlighted by an orange box.

- On the Attach Policy page, select the **AdministratorAccess** policy. Then, click **Next Step**.

Create Role

- Step 1: Set Role Name
- Step 2: Select Role Type
- Step 3: Establish Trust
- Step 4: Attach Policy**
- Step 5: Review

Attach Policy

Select one or more policies to attach. Each role can have up to 10 policies attached.

Filter: Policy Type Filter Showing 187 results

	Policy Name	Attached Entities	Creation Time	Edited Time
☒	AdministratorAccess	1	2015-02-07 00:09 UTC+0530	2015-02-07 00:09 UTC+0530
☐	AmazonAPIGatewayAdministr...	0	2015-07-09 23:04 UTC+0530	2015-07-09 23:04 UTC+0530
☐	AmazonAPIGatewayInvokeFul...	0	2015-07-09 23:06 UTC+0530	2015-07-09 23:06 UTC+0530
☐	AmazonAPIGatewayPushToCL...	0	2015-11-12 05:11 UTC+0530	2015-11-12 05:11 UTC+0530
☐	AmazonAppStreamFullAccess	0	2015-02-07 00:10 UTC+0530	2015-02-07 00:10 UTC+0530
☐	AmazonAppStreamReadOnlyA...	0	2015-02-07 00:10 UTC+0530	2015-02-07 00:10 UTC+0530
☐	AmazonCognitoDeveloperAuth...	0	2015-03-24 22:52 UTC+0530	2015-03-24 22:52 UTC+0530
☐	AmazonCognitoPowerUser	0	2015-03-24 22:44 UTC+0530	2015-03-24 22:44 UTC+0530
☐	AmazonCognitoReadOnly	0	2015-03-24 22:36 UTC+0530	2015-03-24 22:36 UTC+0530

Cancel Previous **Next Step**

7. On the **Review** page, review the details of your assigned role. Finally, click **Create Role**.

Create Role

- Step 1: Set Role Name
- Step 2: Select Role Type
- Step 3: Establish Trust
- Step 4: Attach Policy
- Step 5: Review**

Review

Review the following role information. To edit the role, click an edit link, or click **Create Role** to finish.

Role Name Role1_Lead [Edit Role Name](#)

Role ARN arn:aws:iam::183797070527:role/Role1_Lead

Trusted Entities The account 183797070527

Policies arn:aws:iam::aws:policy/AdministratorAccess [Change Policies](#)

Give this link to users who can switch roles in the console https://signin.aws.amazon.com/switchrole?account=183797070527&roleName=Role1_Lead [Copy Link](#)

Cancel Previous **Create Role**

8. A Role is created.

The screenshot shows the AWS IAM console interface. On the left is a navigation sidebar with links to Dashboard, Search IAM, Details, Groups, Users, Roles (highlighted), Policies, Identity Providers, Account Settings, Credential Report, and Encryption Keys. The main content area is titled 'Create New Role' and 'Role Actions'. It features a search bar and a table of roles. The table has two columns: 'Role Name' and 'Creation Time'. Two roles are listed: 'aws-elasticbeanstalk-ec2-role' and 'Role1_Lead'. The 'Role1_Lead' row is highlighted with an orange border. The footer contains 'Feedback', 'English', copyright information, and links to 'Privacy Policy' and 'Terms of Use'.

Role Name	Creation Time
aws-elasticbeanstalk-ec2-role	2016-01-28 13:41 UTC+0530
Role1_Lead	2016-01-30 16:05 UTC+0530